DELHIPUBLICSCHOOLJANKIPURAM,LUCKNOW

SESSION2024-2025: PERIODIC ASSESSMENT-II

REVISION SHEET

20

Subject:SCIENCE

Date: 10/9/24 Name : Md. Hifzaan Khan Class/Sec: IX D

Choose the correct option:

1. If the temperature of an object is 258K, it will be equivalent to:
a) -15°C
b) 15°C
c) 368°C
d) -25°C

2. During summer days, water kept in an earthen pot (pitcher) becomes cool because of thephenomenon of:
a) Diffusion
b) Transpiration
c) Osmosis
d) Evaporation

3. The rusting of an iron object is called:
a) Corrosion and it is a physical as well as a chemical changes.
b) Dissolution and it is a physical change.
c) Corrosion and it is a chemical change.
d) Dissolution and it is a chemical change

4. Which of the following is a suspension?
a) Milk
b) Milk of magnesia
c) Salt Solution
d) Vinegar

5. In all the three states of water, (i.e. ice, liquid and vapour) chemical composition of water
a) is very different
b) remains same
c) sometimes same and sometimes different
d) none of the above

6. What is true about homogeneous mixture?
a) Homogeneous mixture is the mixture of two or more than two components.
b) In homogeneous mixture the composition and properties are uniform throughout the mixture
c) both (a) and (b) are true
d) none of the above

7. Which of the following properties does not describe a compound?

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a) It is composed of two or more elements
b) It is a pure substance.
c) It cannot be separated into constituents by physical means
d) It is mixed in any proportion by mass

8. In the tincture of iodine, find the solute and solvent?
a) alcohol is the solute and iodine is the solvent
b) iodine is the solute and alcohol is the solvent
c) any component can be considered as solute or solvent
d) tincture of iodine is not a solution

Direction: In the following questions(Q 9 to 11), there are two statements marked as Assertion and Reason. Mark your answer as per the codes provided below:
(a)Both Assertion and Reason are true and Reason is the correct explanation of Assertion.
(b)Both Assertion and Reason are true but Reason is not the correct explanation of Assertion.
(c) Assertion is true but Reason is false.
(d) Assertion is false but Reason is true.

9. Assertion: When a beam of light is passed through a colloidal solution placed in a dark place, the path of the beam becomes visible. Reason: light gets scattered by the colloidal particles.

10. Assertion: When a solid melts, its temperature remains the same.
Reason: The heat gets used up in changing the state by overcoming the forces of attraction between the particles.

11. Assertion: A solution of table salt in a glass of water is homogeneous.
Reason: A solution having different composition throughout is homogeneous.

VERY SHORT ANSWER TYPE QUESTIONS (2 MARKS)

12. A small volume of water in a kettle can fill a kitchen with steam. Explain.

13. You have been given a suspension and a true solution. How will you differentiate between them by their appearance?

14. Differentiate between boiling and evaporation.

OR

On a hot, humid day, a student noticed droplets of water forming on the outside of a glass of cold water. Explain the phenomenon.

15. Is water an element or a compound? Give two points in support of your statement.

SHORT ANSWER TYPE QUESTIONS (3 MARKS)

16. When water is cooled to a temperature X, it is converted into ice at temperature X by a process called P.
a) What is the value of temperature X in Kelvin?
b) What is the process P known as?
c) What is the name of energy released during process P?

17. Analyse the characteristics of water that classify it as a compound rather than a mixture, providing at least three specific reasons to support your explanation.

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18. (i) Classify the following as physical or chemical changes: -
(a) Burning of magnesium wire
(b) Freezing of water
(c) Tearing of paper
(d) Glowing of an electric bulb

(ii) Calculate the concentration of solution, which contains 2.5g of salt dissolved in 50 g of water.

(iii) You are given two beakers with the same amount of water. In one beaker, you dissolve 10 grams of sugar at room temperature, and in the other, you dissolve 20g of sugar. Which solution will have a higher concentration and why?

OR

100 ml of water at room temperature of 25°C is taken in a beaker and a little of solid S is dissolved in it by stirring to obtain a solution X. More and more of solid S is added to the solution with constant stirring, while keeping the temperature of solution constant at 30°C. After some time it is observed that no more solid dissolves in water and at the same time some solid is also left undissolved at the bottom of the beaker. The contents of beaker are filtered through a filter paper to obtain solution Y in the form of a filtrate.

(a) Analyse the properties of solution X & Y and identify the specific type of solution itrepresents.
(b) What will you observe if the solution Y at 30°C is cooled down to 10°C by keeping thebeaker in crushed ice? Why?
(c) What term is used to denote the amount of solid dissolved in 100 g of water in a solutionlike Y?

19. The temperature-time graph given alongside shows the heating curve for pure wax.From the graph answer the following:
(a) What is the physical state of the substance at the points A, B, C and D?
(b) What is the melting point of the substance?
(c) What is its boiling point?
(d) Which portions of the graph indicates that change of state is taking place?

OR

Discuss four factors which affect the rate of evaporation. Latent heat of evaporation of two liquids A and B is 100 J/kg and 150 J/kg respectively. Which one can produce more cooling effect and why?

20. A solution contains 30 g of glucose, 20 g of salt in 500 mL of water. Calculate the mass per cent of
(a) glucose, (b) salt (density of water = 1 g /mL).

21.(i)Identify the dispersed phase and dispersing medium in the following colloids.
(a) Fog
(b) Cheese
(c) Coloured gemstone

(ii) On heating, calcium carbonate gets converted into calcium oxide and carbon dioxide.
(a) Is this a physical or a chemical change?
(b) Can you prepare one acidic and one basic solution by using the products formed in the above process? If so, write the chemical equation involved.

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CASE BASED AND SOURCE BASED QUESTIONS (4 MARKS)

22. When water is heated to a temperature X, it gets converted into steam at temperature X by a processcalled R. And when steam at temperature X is cooled, it gets reconverted into water at the same temperatureX by a process called S.

(i) How much is the value of X in Kelvin?

(ii) What is the process R called and define it also.

(iii) What is the name of the energy absorbed during the process R?

OR

What is the name of energy released during the process S known as?

23. In an experimental activity, crushed ice was taken in a beaker. A thermometer is fitted in such a way that its bulb was thoroughly surrounded by ice. The beaker is now slowly heated and temperature was regularly noted. Temperature rises gradually as the heating is continued and becomes constant when ice starts changing intoliquid. Select the correct answers for the following questions:

i) What name is associated with conversion of ice into water?

a. Evaporation
b. Sublimation
c. Freezing
d. Fusion of Solid

ii) What specific name is given to the constant temperature?

a. latent heat of fusion
b. Boiling Point
c. Melting Point
d. Condensation point

iii) The heat added to the system at constant temperature is called

a) specific heat
b) latent heat
c) residual heat
d) none of the above

iv) Where does the heat energy go when the temperature does not rise?

a. It makes the molecular motion of the liquid faster
b. It raises the temperature of the beaker only.
c. It is utilised for bringing out the complete change of state
d. It slows down the molecular motion

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